PROJECT BRIEF / FULL-STACK ENGINEERING

CentraLog

Connecting asset custody with maintenance and valuation.

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PROBLEM & USERS

Institutions need to know where equipment is, who holds it, and whether it is in use or under repair. CentraLog connects physical tracking with a valuation calculation that uses maintenance history.

Built for: Inventory staff · managers · accountants · administrators

MY CONTRIBUTION

I built CentraLog independently as a school project, implementing the asset lifecycle, backend services, frontend workflows, and valuation strategies.

MAIN TECHNOLOGIES

ASP.NET Core 8 / C# / React 19 / TypeScript / EF Core 8 / MySQL / MariaDB / JWT / Vitest / xUnit

VERSION SCOPE

Reviewed asset services and valuation strategies. Code references show the asset service and strategy calculations at fixed commits. The portfolio calculator and lifecycle walkthrough run independently using sample data.

WORKFLOW

01. Register and activate

02. Move with a record

03. Track maintenance

04. Calculate and report

SYSTEM STRUCTURE

Interface: React / TypeScript

API: ASP.NET Core / JWT

Domain: Depreciation strategies / worker

Persistence: EF Core / MySQL or MariaDB

ENGINEERING DECISION

Use interchangeable depreciation strategies to calculate values from asset and maintenance records on read. Group multi-asset mutations and audit entries inside EF Core transactions.

TRADE-OFF

Calculating on read adds computation but avoids maintaining a separate cached valuation. Repair windows are handled through the application's explicitly defined downtime-adjustment policy.

READ CASE STUDY | TRY WALKTHROUGH | VIEW SOURCE

Portfolio case studies and walkthroughs provide further detail and sample interactions.